

AZ-900 Practice Questions

Question 1

A company is migrating from an on-premises datacenter to Azure. The IT director wants to understand how costs will change with the migration. Currently, the company pays fixed monthly costs for their datacenter regardless of actual resource usage. Which statement best describes how the consumption-based model in Azure will change their cost structure?

- **✗ The company will pay the same total costs as their on-premises datacenter, but distributed across multiple Azure subscriptions.**
 - This answer is incorrect. The consumption-based model does not guarantee the same total costs as on-premises infrastructure. Costs in Azure depend entirely on actual resource consumption and can be higher or lower than on-premises costs depending on usage patterns, optimization efforts, and resource requirements, not simply distributed differently.
- **✗ The company will pay upfront capital expenses for Azure infrastructure before they can deploy any resources.**
 - This answer is incorrect. The consumption-based model eliminates the need for upfront capital expenses on infrastructure. Instead of purchasing physical hardware before use, Azure allows organizations to consume resources on-demand and pay for them as operational expenses, removing the capital expenditure barrier to entry.
- **✓ The company will pay only for the Azure resources they actually use, with costs varying based on consumption.**
 - This answer is correct. The consumption-based model means the company pays only for the resources they actually consume, with costs fluctuating based on usage levels. This represents a shift from the fixed operational expenses of their on-premises datacenter to variable operational expenses that scale with actual resource consumption, providing cost flexibility and efficiency.
- **✗ The company will pay a fixed monthly subscription fee for Azure services, similar to their current datacenter costs.**
 - This answer is incorrect. Azure's consumption-based model does not charge fixed monthly subscription fees for most services. While some services offer

reserved instances or commitments, the fundamental characteristic of the consumption-based model is that costs vary based on actual usage, not fixed monthly fees regardless of consumption.

Overall explanation: The consumption-based model is a fundamental cloud computing concept where customers pay only for the resources they consume, converting IT costs from capital expenses (CapEx) to operational expenses (OpEx). This model provides flexibility, eliminates waste from paying for unused capacity, and allows costs to scale dynamically with business needs, representing a significant departure from traditional fixed-cost datacenter models.

Question 2

A retail company runs an e-commerce web application on Azure. During normal business hours, the application serves approximately 1,000 users. However, during seasonal sales events, traffic increases to 50,000 concurrent users. The company wants to ensure the application can handle these traffic spikes while minimizing costs during normal periods. Which cloud capability best addresses this requirement?

-  **Horizontal scaling (scaling out)**
 - This answer is correct. Horizontal scaling (scaling out) involves adding more instances of resources, such as virtual machines, when demand increases and removing them when demand decreases. This approach perfectly addresses the scenario by allowing the application to automatically add servers during the 50,000-user sales events and scale back down during normal 1,000-user periods, optimizing both performance and cost.
-  **High availability**
 - This answer is incorrect. High availability refers to ensuring services remain operational and accessible even during failures by using redundancy and failover mechanisms. While high availability is important for business continuity, it does not specifically address the need to dynamically adjust capacity based on fluctuating user demand or optimize costs during different traffic periods.
-  **Geo-distribution**
 - This answer is incorrect. Geo-distribution involves deploying resources across multiple geographic regions to reduce latency for users in different locations and provide disaster recovery capabilities. While this can improve performance for

global users, it does not directly address the requirement to dynamically adjust capacity for varying user loads or minimize costs during normal operations.

- **✗ Vertical scaling (scaling up)**
 - This answer is incorrect. Vertical scaling involves increasing the resources (CPU, RAM) of existing virtual machines, which has practical limits and typically requires downtime. While vertical scaling can handle some growth, it is not the most cost-effective solution for handling dramatic traffic spikes from 1,000 to 50,000 users, as you would need to maintain expensive high-capacity resources even during normal periods.

Overall explanation: Scalability in the cloud refers to the ability to adjust resources to meet changing demand. Horizontal scaling (scaling out/in) is the most effective approach for handling large variations in traffic because it allows automatic addition or removal of resource instances based on demand, ensuring optimal performance during peak times while minimizing costs during normal periods by running only the necessary resources.

Question 3

Your company has two Azure virtual networks: VNet-Production in the East US region and VNet-Development in the West Europe region. The development team needs to enable direct communication between virtual machines in both virtual networks using private IP addresses. Which Azure networking solution should you implement?

- **✗ Create a shared subnet that spans both virtual networks**
 - This answer is incorrect. Subnets exist within a single virtual network and cannot span across multiple virtual networks. Each virtual network maintains its own address space and subnets, so creating a shared subnet between two separate virtual networks is not architecturally possible in Azure.
- **✓ Configure virtual network peering between VNet-Production and VNet-Development**
 - This answer is correct. Virtual network peering is specifically designed to connect Azure virtual networks, enabling resources in different virtual networks to communicate with each other using private IP addresses. Peering works across regions (global VNet peering), provides low-latency and high-bandwidth connectivity, and is the recommended Azure-native solution for this scenario.

-  **Deploy Azure DNS to enable name resolution between the two virtual networks**
 - This answer is incorrect. Azure DNS provides name resolution services for domain names and does not establish network connectivity between virtual networks. While Azure DNS can help resolve names after connectivity is established, it does not enable the actual network communication between VMs in different virtual networks.
-  **Configure Azure VPN Gateway to establish a site-to-site VPN connection between the two virtual networks**
 - This answer is incorrect. While Azure VPN Gateway can connect virtual networks, it is primarily designed for connecting on-premises networks to Azure or for point-to-site connections. For connecting two Azure virtual networks, VPN Gateway would work but is more complex, costly, and provides lower bandwidth compared to the more appropriate solution of virtual network peering.

Overall explanation: Virtual network peering is the Azure service specifically designed to connect virtual networks together, whether in the same region or across different regions. It provides secure, high-performance connectivity using Microsoft's private backbone network, allowing resources to communicate using private IP addresses as if they were in the same network. This is the most efficient and cost-effective solution for connecting Azure virtual networks compared to alternatives like VPN Gateway.

Question 4

Your organization is deploying a web application to Azure. The application must be accessible to users worldwide over the internet without requiring a VPN connection or private network configuration. Which cloud deployment model best describes this accessibility requirement?

-  **Private cloud**
 - This answer is incorrect. A private cloud is dedicated exclusively to a single organization and is typically hosted on-premises or in a dedicated datacenter. Private clouds require special network configurations such as VPNs or dedicated connections for access, which contradicts the requirement for worldwide internet accessibility without VPN.
-  **Public cloud**

- This answer is correct. Public cloud services like Azure provide resources over the public internet, making them accessible to anyone with internet connectivity worldwide. This deployment model aligns perfectly with the requirement for global accessibility without VPN connections, as public cloud resources are designed to be reached through standard internet protocols.
- **✗ Hybrid cloud**
 - This answer is incorrect. A hybrid cloud combines both private and public cloud elements, typically connecting on-premises infrastructure with cloud services. While hybrid clouds can include publicly accessible components, the hybrid model itself emphasizes the integration of private and public resources rather than purely public internet accessibility.
- **✗ Community cloud**
 - This answer is incorrect. A community cloud is shared by several organizations with common concerns (security, compliance, jurisdiction) and may be managed internally or by a third party. Community clouds are not a standard deployment model emphasized in Azure fundamentals and don't specifically address the requirement for unrestricted worldwide internet access.

Overall explanation: The public cloud deployment model is characterized by resources and services that are owned and operated by a third-party cloud provider and delivered over the public internet. Azure, as a public cloud platform, enables applications to be accessible globally through internet connectivity without requiring private network configurations like VPNs. This makes public cloud the ideal choice for applications requiring worldwide accessibility.

Question 5

Your organization has deployed multiple Azure resources across several departments including Marketing, Finance, and IT. The CFO needs to generate monthly cost reports that break down Azure spending by department. What is the most effective way to accomplish this requirement?

- **✗ Use Microsoft Defender for Cloud to track and report on departmental resource costs.**
 - This answer is incorrect. Microsoft Defender for Cloud is a security posture management and threat protection service, not a cost management or billing tool. While it provides security recommendations and compliance monitoring, it does

not have features designed for tracking or reporting on resource costs by department.

- ☒ **Apply tags with department names to resources and use cost analysis to filter and group spending by tag values.**
 - This answer is correct. Tags are name-value pairs that can be applied to Azure resources specifically for organizational purposes such as cost tracking, resource management, and categorization. Azure Cost Management natively supports filtering and grouping by tags, making this the most efficient and scalable solution for tracking departmental spending without restructuring the Azure environment.
- ☐ **Create separate resource groups for each department and manually calculate costs per resource group.**
 - This answer is incorrect. Although organizing resources into resource groups by department provides some logical organization, Azure Cost Management does not automatically break down costs by resource group in the same flexible way it does with tags. Additionally, manual cost calculation is inefficient and error-prone when Azure provides automated cost analysis tools that work seamlessly with tags.
- ☐ **Create separate Azure subscriptions for each department and generate cost reports per subscription.**
 - This answer is incorrect. While creating separate subscriptions per department would enable cost tracking, it is an overly complex solution that introduces unnecessary administrative overhead and does not leverage Azure's built-in cost management capabilities. Subscriptions are meant for larger organizational boundaries and compliance requirements, not for simple cost allocation scenarios.

Overall explanation: Tags are metadata elements designed specifically for organizing, managing, and tracking Azure resources across various dimensions including cost allocation, environment classification, and ownership. The primary purpose of tags in cost management scenarios is to enable flexible filtering and grouping in Azure Cost Management and billing reports without requiring structural changes to the Azure environment such as creating new subscriptions or resource groups.

Question 6

Your organization uses Microsoft Entra ID to manage employee identities. You need to allow users from a partner company to access a specific application in your Azure environment without creating separate user accounts for them in your directory. The partner company will manage their users' credentials and lifecycle. Which Microsoft Entra feature should you use?

-  **Microsoft Entra B2B collaboration**
 - This answer is correct. Microsoft Entra B2B (business-to-business) collaboration allows you to invite external users from partner organizations to access your applications and resources as guest users. The partner organization maintains control over their users' identities and credentials, while you control access to your resources, making this the ideal solution for the scenario.
-  **Microsoft Entra B2C**
 - This answer is incorrect. Microsoft Entra B2C (business-to-consumer) is designed for customer-facing applications where you need to manage authentication for consumers using social, enterprise, or local account identities. This scenario involves business partner collaboration, not consumer identity management, making B2B the appropriate choice instead of B2C.
-  **Microsoft Entra Conditional Access**
 - This answer is incorrect. Microsoft Entra Conditional Access is a security feature that enforces access policies based on conditions such as user location, device state, or risk level. While it can be used alongside B2B collaboration to secure external access, it is not the feature that enables external users from partner organizations to access your resources in the first place.
-  **Microsoft Entra Domain Services**
 - This answer is incorrect. Microsoft Entra Domain Services provides managed domain services such as domain join, group policy, and LDAP, which are typically used for lifting and shifting on-premises applications to Azure. It does not address the requirement of allowing external partner users to access applications while their organization manages their credentials.

Overall explanation: Microsoft Entra B2B collaboration is specifically designed for business-to-business scenarios where external users need access to your organization's resources. It allows partner organizations to maintain ownership of their users' identities while

enabling secure, controlled access to your applications through guest user accounts. This eliminates the administrative overhead of creating and managing separate accounts for external users in your directory.

Question 7

A retail company is planning to migrate their on-premises application to Azure. The finance department needs to forecast their annual cloud spending accurately for budget planning. The IT manager wants to ensure they can predict monthly costs within a 5% variance. Which Azure benefit of cloud computing best addresses this requirement?

-  **High availability**
 - This answer is incorrect. High availability refers to ensuring that services and applications remain accessible and operational with minimal downtime, typically measured in uptime percentages. While high availability is valuable for business continuity, it does not directly help with forecasting or controlling cloud spending for budget planning purposes.
-  **Performance predictability**
 - This answer is incorrect. Performance predictability refers to the ability to predict and expect consistent performance levels for workloads, such as knowing that applications will respond within specific timeframes or handle certain loads. While important for application reliability, it does not address the finance department's need to forecast cloud spending accurately.
-  **Cost predictability**
 - This answer is correct. Cost predictability is the cloud benefit that enables organizations to forecast and control their cloud spending through tools like Azure Cost Management, pricing calculators, and cost alerts. This benefit allows the finance department to accurately predict monthly and annual expenses, helping them stay within budget parameters like the 5% variance mentioned in the scenario.
-  **Disaster recovery**
 - This answer is incorrect. Disaster recovery involves planning and implementing strategies to restore services and data after catastrophic events or failures.

Although disaster recovery is crucial for business resilience, it does not provide capabilities for predicting or managing cloud costs for financial forecasting.

Overall explanation: Predictability in the cloud encompasses two main aspects: performance predictability and cost predictability. In this scenario, the finance department's requirement to forecast annual spending and maintain costs within a specific variance directly aligns with cost predictability, which Azure enables through various tools and consumption-based pricing models that allow organizations to plan and control their cloud expenditures effectively.

Question 8

A retail company is migrating its e-commerce application to Azure. The IT director is concerned about protecting customer data and maintaining compliance with industry regulations. The director asks you to explain how Azure's built-in security features can help reduce the company's security management burden compared to their on-premises data center. Which benefit of Azure security best addresses this concern?

-  **Azure eliminates all security responsibilities from the customer organization by handling all security aspects of the cloud environment.**
 - This answer is incorrect. Azure follows a shared responsibility model where security responsibilities are divided between Microsoft and the customer. While Microsoft secures the cloud infrastructure, customers remain responsible for securing their data, applications, identities, and access management, so not all security responsibilities are eliminated from the customer.
-  **Azure provides security tools and features that are managed and updated by Microsoft, reducing the organization's responsibility for maintaining security infrastructure.**
 - This answer is correct. Azure implements a shared responsibility model where Microsoft manages security of the underlying cloud infrastructure, including physical datacenters, networks, and hosts, while also providing built-in security tools like Microsoft Defender for Cloud and continuous security updates. This significantly reduces the customer's burden of maintaining security infrastructure compared to on-premises environments where the organization must handle all security layers.
-  **Azure provides security only for Microsoft-managed services and requires third-party solutions for customer-deployed resources.**

- This answer is incorrect. Azure provides comprehensive security features across both Microsoft-managed services and customer-deployed resources. Services like Microsoft Defender for Cloud monitor and protect customer workloads including virtual machines, databases, and applications, not just Microsoft-managed platform services.
- **✗ Azure requires customers to install and maintain their own security software on all virtual machines to ensure compliance.**
 - This answer is incorrect. Azure provides numerous built-in security features and tools, such as Microsoft Defender for Cloud, Microsoft Entra ID, and Azure DDoS Protection, which customers can leverage without installing separate third-party security software. While customers can add additional security solutions if needed, Azure does not require customers to install and maintain their own security software for basic protection.

Overall explanation: The key benefit of Azure security is the shared responsibility model, which reduces the customer's security management burden by having Microsoft handle infrastructure security, physical security, and platform updates. This allows organizations to focus on application-level security and data protection rather than maintaining the entire security stack as they would in an on-premises environment. Understanding this division of responsibility is fundamental to AZ-900 knowledge about cloud security benefits.

Question 9

Your organization has 500 employees who need to access both Microsoft 365 applications and custom applications running in Azure. The IT department wants to implement a centralized identity solution where users can sign in once and access all applications without re-authenticating. Which Azure service should you recommend?

- **✗ Windows Server Active Directory running on Azure Virtual Machines**
 - This answer is incorrect. Windows Server Active Directory running on Azure VMs is an infrastructure-as-a-service solution that requires management overhead and is primarily suited for on-premises-style domain services. It does not natively provide SSO for Microsoft 365 and modern cloud applications without additional integration with Microsoft Entra ID.
- **✗ Azure Active Directory Domain Services (Azure AD DS)**

- This answer is incorrect. While Azure Active Directory Domain Services provides managed domain services like domain join and group policy, it is primarily designed for lift-and-shift scenarios requiring traditional domain services (LDAP, Kerberos, NTLM) for legacy applications. It is not the primary solution for implementing SSO across Microsoft 365 and modern Azure applications.
-  **Microsoft Entra ID**
 - This answer is correct. Microsoft Entra ID is Microsoft's cloud-based identity and access management service that provides single sign-on (SSO) capabilities for both Microsoft 365 and Azure applications. It serves as the centralized identity provider for cloud and hybrid environments, enabling users to authenticate once and access multiple applications seamlessly.
-  **Azure Key Vault**
 - This answer is incorrect. Azure Key Vault is designed to securely store and manage secrets, keys, and certificates used by applications and services. It is not an identity and access management service and does not provide authentication or single sign-on capabilities for users.

Overall explanation: Microsoft Entra ID is the foundational identity service in Azure that provides authentication, authorization, and single sign-on capabilities for cloud applications. For this scenario requiring centralized identity management and SSO across Microsoft 365 and Azure applications, Microsoft Entra ID is the appropriate and most efficient solution, as it is purpose-built for cloud identity management.

Question 10

A company is planning to migrate their on-premises infrastructure to Azure. They need to assess their current environment, which includes 50 physical servers running various workloads, multiple databases, and web applications. The IT manager wants to discover dependencies between servers, get Azure cost estimates, and receive right-sizing recommendations before migrating. Which Azure service should they use to accomplish these assessment tasks?

-  **Azure Advisor**
 - This answer is incorrect. Azure Advisor is a personalized cloud consultant that provides best practice recommendations for optimizing existing Azure resources across cost, security, reliability, operational excellence, and performance. It

analyzes resources already deployed in Azure rather than assessing on-premises infrastructure for migration planning purposes.

- **Microsoft Defender for Cloud**

- This answer is incorrect. Microsoft Defender for Cloud is a cloud-native application protection platform that provides security posture management and threat protection for workloads in Azure, on-premises, and multi-cloud environments. While it can assess security configurations, it is not designed for migration assessment, dependency mapping, or providing cost and sizing recommendations for migration projects.

- **Azure Migrate**

- This answer is correct. Azure Migrate is the centralized hub specifically designed for assessing and migrating on-premises servers, databases, and applications to Azure. It provides discovery of on-premises assets, dependency analysis between servers, Azure cost estimates, and right-sizing recommendations to help organizations plan their migration effectively.

- **Azure Monitor**

- This answer is incorrect. Azure Monitor is designed for collecting, analyzing, and acting on telemetry data from Azure and on-premises environments after resources are deployed. While it can track performance and availability, it is not designed for pre-migration assessment, dependency mapping, or providing migration cost estimates and sizing recommendations.

Overall explanation: Azure Migrate is Microsoft's primary solution for migration projects, offering a unified platform for assessment and migration. It provides essential pre-migration capabilities including agentless discovery, application dependency visualization, performance-based sizing recommendations, and cost estimation for running workloads in Azure. These features directly address the company's stated needs for assessing their environment before migration.

Question 11

A company has five branch offices, each with a local Windows Server file server that stores frequently accessed project files. The IT department wants to centralize file storage in Azure Files while ensuring that users at each branch office can access files quickly without

experiencing latency, even when the internet connection is slow. Which Azure service should the company implement to meet these requirements?

- **Azure NetApp Files**

- This answer is incorrect. Azure NetApp Files is an enterprise-grade file storage service suitable for high-performance workloads, but it does not provide local caching capabilities at branch offices. Users would still experience latency when accessing files over the internet, and it doesn't offer the synchronization features needed for this hybrid scenario.

- **Azure Files with geo-replication**

- This answer is incorrect. Azure Files with geo-replication would provide redundancy across Azure regions but does not address the local access performance requirement. Users at branch offices would still need to access files directly over the internet from Azure, which would result in latency issues, especially when internet connections are slow.

- **Azure File Sync**

- This answer is correct. Azure File Sync enables the company to centralize file storage in Azure Files while maintaining local caches of frequently accessed files on each branch office's Windows Server. This provides fast local access to files while synchronizing changes bidirectionally between on-premises servers and Azure Files, effectively solving both the centralization and performance requirements.

- **Azure Blob Storage with Content Delivery Network (CDN)**

- This answer is incorrect. While Azure Blob Storage with CDN can cache content at edge locations for faster access, it is designed for web content delivery and does not provide bidirectional synchronization with on-premises Windows file servers. Users would not be able to access files through standard Server Message Block (SMB) file shares as they typically would with file servers.

Overall explanation: Azure File Sync is specifically designed for hybrid scenarios where organizations need to centralize file storage in Azure while maintaining local performance at branch offices. It creates a cloud tiering system that keeps frequently accessed files cached locally on Windows Servers while storing the complete file set in Azure Files, with automatic bidirectional synchronization ensuring consistency across all locations.

Question 12

A company is developing a mobile application that needs to process image uploads from users. The image processing workload is unpredictable, with usage spikes occurring randomly throughout the day. The company wants to minimize costs by only paying for compute resources when images are actually being processed. Which Azure compute solution best meets these requirements?

-  **Azure Functions with a consumption plan**
 - This answer is correct. Azure Functions with a consumption plan is a serverless solution that automatically scales based on demand and charges only for the actual execution time and resources consumed. When no images are being processed, there are no compute charges, making it ideal for unpredictable workloads with the pay-per-execution pricing model.
-  **Azure Virtual Machines with auto-scaling enabled**
 - This answer is incorrect. While Azure Virtual Machines with auto-scaling can adjust capacity based on demand, the company still pays for VMs even during low-usage periods, and there is a minimum number of instances that must run. This does not align with the serverless model of paying only for actual execution time.
-  **Azure App Service with an Always On feature**
 - This answer is incorrect. Azure App Service with the Always On feature keeps the application constantly running and loaded in memory, which means the company pays for continuous compute resources regardless of actual usage. This contradicts the requirement to pay only when processing occurs.
-  **Azure Container Instances running continuously**
 - This answer is incorrect. Azure Container Instances running continuously would incur charges for the entire time the containers are running, regardless of whether images are being processed. This does not provide the cost optimization of paying only when work is being performed.

Overall explanation: Serverless computing in Azure, exemplified by Azure Functions with a consumption plan, provides automatic scaling and a pay-per-execution pricing model where customers are charged only for the resources consumed during actual code execution. This model is ideal for unpredictable workloads because it eliminates charges during idle periods

while automatically handling demand spikes. The serverless approach removes infrastructure management responsibilities and optimizes costs for event-driven scenarios like image processing.

Question 13

A retail company currently manages its own data center with physical servers that often sit idle during off-peak hours. The company pays fixed costs for hardware, cooling, and maintenance regardless of actual usage. They are considering moving to Azure to optimize their IT spending. Which characteristic of cloud computing would most directly address their concern about paying for unused capacity?

-  **High availability**
 - This answer is incorrect. High availability refers to the ability to keep services running with minimal downtime, typically through redundancy and failover mechanisms. While high availability is an important cloud characteristic, it does not address the concern about paying for unused capacity during off-peak hours.
-  **Disaster recovery**
 - This answer is incorrect. Disaster recovery involves the ability to recover systems and data after a catastrophic event. Although cloud platforms provide enhanced disaster recovery capabilities, this characteristic does not address the specific concern about optimizing costs related to idle infrastructure during low-usage periods.
-  **Geo-distribution**
 - This answer is incorrect. Geo-distribution refers to the ability to deploy resources across multiple geographic regions for better performance and compliance. While this is a valuable cloud feature, it does not solve the problem of paying for unused capacity or idle resources during off-peak periods.
-  **Pay-as-you-go pricing**
 - This answer is correct. Pay-as-you-go pricing, also known as consumption-based pricing, is a fundamental characteristic of cloud computing that allows organizations to pay only for the resources they actually use. This directly addresses the company's concern about paying fixed costs for idle servers, as

they can scale resources down during off-peak hours and only pay for what they consume.

Overall explanation: Cloud computing's consumption-based pricing model is a core characteristic that distinguishes it from traditional on-premises infrastructure. Instead of paying fixed capital expenses for hardware that may sit idle, organizations using cloud services pay only for the compute resources, storage, and services they actually consume, allowing them to align costs with actual business demand and optimize their IT spending.

Question 14

A company needs to store large amounts of unstructured data including images, videos, and log files for their web application. The data will be accessed frequently during the first month after upload, but afterward will rarely be accessed. The company wants to minimize storage costs while ensuring the data remains available if needed. Which Azure storage solution should they implement?

- ☒ **Azure Blob Storage with a lifecycle management policy to move data to the Cool access tier after 30 days**
 - This answer is correct. Azure Blob Storage is specifically designed for storing unstructured data such as images, videos, and log files. Lifecycle management policies can automatically transition blobs from the Hot tier (optimized for frequent access) to the Cool tier (optimized for infrequent access with lower storage costs) after 30 days, perfectly matching the company's access patterns and cost optimization requirements.
- ☐ **Azure Queue Storage with automatic archiving enabled**
 - This answer is incorrect. Azure Queue Storage is designed for storing and retrieving messages for asynchronous communication between application components, not for storing large unstructured data files like images and videos. Queue Storage does not have archiving capabilities and is not an appropriate solution for this scenario.
- ☐ **Azure Files with Premium performance tier for the first month, then switch to Standard tier**
 - This answer is incorrect. Azure Files is designed for file shares that can be accessed via SMB or NFS protocols, typically used for shared application data or replacing on-premises file servers. While it can store unstructured data, it lacks

the lifecycle management capabilities for automatic tier transitions, and it is not optimized for the large-scale object storage scenario described in the question.

-  **Azure Table Storage with partitioning by upload date**
 - This answer is incorrect. Azure Table Storage is a NoSQL key-value store designed for storing structured data with a schema, such as application configuration data or user profiles. It is not optimized for storing large unstructured files like images, videos, and log files, making it unsuitable for this scenario.

Overall explanation: Azure Blob Storage is the primary Azure service for storing unstructured data at scale, supporting various data types including images, videos, and logs. The lifecycle management feature allows automatic cost optimization by transitioning data between access tiers (Hot, Cool, and Archive) based on defined rules, making it ideal for scenarios where access patterns change over time. This combination of capabilities directly addresses both the technical requirements and cost optimization goals outlined in the scenario.

Question 15

A company is developing a new web application and needs to deploy it to Azure. The development team wants to focus on writing code without managing the underlying infrastructure, operating system, or web server configuration. The solution must support automatic scaling based on traffic and provide built-in load balancing. Which Azure service should the company use?

-  **Azure Virtual Machines**
 - This answer is incorrect. Azure Virtual Machines require the company to manage the underlying operating system, web server configuration, and infrastructure, which contradicts the requirement to avoid infrastructure management. While VMs provide full control, they require significantly more administrative overhead than a Platform-as-a-Service (PaaS) solution.
-  **Azure App Service**
 - This answer is correct. Azure App Service is a fully managed Platform-as-a-Service (PaaS) that allows developers to deploy web applications without managing infrastructure, operating systems, or web servers. It provides built-in automatic scaling, load balancing, and supports multiple programming

languages, making it ideal for this scenario where the team wants to focus solely on application code.

- **✗ Azure Container Instances**

- This answer is incorrect. While Azure Container Instances provides a way to run containers without managing infrastructure, it is primarily designed for simple containerized applications and background tasks, not full-featured web applications. Additionally, it requires the application to be containerized and does not provide the same level of built-in web hosting features, automatic scaling, and load balancing as a dedicated web application platform.

- **✗ Azure Functions**

- This answer is incorrect. Azure Functions is a serverless compute service designed for event-driven, short-running code execution rather than hosting full web applications. While it eliminates infrastructure management and supports automatic scaling, it is optimized for specific functions or microservices rather than serving as a complete web application hosting platform with continuous HTTP endpoints.

Overall explanation: Azure App Service is the most appropriate solution for hosting web applications when developers want to focus on code rather than infrastructure management. As a PaaS offering, it abstracts away operating system and web server management while providing enterprise-grade features like automatic scaling, load balancing, deployment slots, and continuous integration support, making it the ideal choice for traditional web application hosting scenarios.

Question 16

Your company wants to improve security for users accessing Microsoft Entra ID-protected resources. The IT director has asked you to recommend an authentication method that eliminates the need for users to remember and enter passwords while still providing strong authentication. Which authentication method should you recommend?

- **✗ Multi-factor authentication using SMS text messages**

- This answer is incorrect. While multi-factor authentication using SMS text messages does provide an additional layer of security beyond passwords, it still requires users to have and remember a password as the first authentication

factor. SMS-based MFA is a supplementary security measure, not a passwordless solution.

- **Windows Hello for Business**

- This answer is correct. Windows Hello for Business is a passwordless authentication method that uses biometric authentication (such as facial recognition or fingerprint) or a PIN tied to a specific device, eliminating the need for traditional passwords. It provides strong, phishing-resistant authentication while improving user experience by removing the burden of password management.

- **Username and complex password with regular password rotation**

- This answer is incorrect. Username and complex password with regular password rotation is a traditional password-based authentication method that requires users to remember and manage passwords. This approach actually increases password management burden and does not meet the requirement for a passwordless solution.

- **Security questions and answers**

- This answer is incorrect. Security questions and answers are a legacy authentication mechanism that still relies on knowledge-based factors users must remember. This method is generally considered less secure than modern authentication options and is not a passwordless solution in the context of modern identity security.

Overall explanation: Passwordless authentication methods eliminate the need for users to remember and enter traditional passwords while maintaining or improving security. Windows Hello for Business, Microsoft Authenticator app, and FIDO2 security keys are examples of passwordless authentication options supported by Microsoft Entra ID. These solutions reduce security risks associated with password-based attacks while improving user experience and productivity.

Question 17

A company is deploying a business-critical application in Azure and needs to protect against datacenter-level failures within a single Azure region. The application requires 99.99% availability SLA. Which Azure feature should the company use to distribute their virtual machines across physically separate locations within the region?

-  **Deploy virtual machines across availability zones**
 - This answer is correct. Availability zones are physically separate datacenters within an Azure region, each with independent power, cooling, and networking. Deploying virtual machines across availability zones protects against datacenter-level failures while keeping resources within the same region, and this configuration supports the 99.99% SLA requirement for virtual machines.
-  **Deploy virtual machines in a single datacenter with multiple backup copies**
 - This answer is incorrect. Azure does not offer an option to deploy virtual machines in a single datacenter with multiple backup copies as a high availability feature. This approach would not protect against datacenter-level failures since all resources would be in the same physical location, making them vulnerable to the same infrastructure issues.
-  **Deploy virtual machines across multiple Azure regions**
 - This answer is incorrect. While deploying across multiple Azure regions does provide high availability and disaster recovery capabilities, it addresses region-level failures rather than datacenter-level failures within a single region. This solution would be more complex and costly than necessary for the stated requirement of protecting against datacenter-level failures within one region.
-  **Deploy virtual machines across multiple resource groups**
 - This answer is incorrect. Resource groups are logical containers for organizing and managing Azure resources, not physical separation mechanisms. Deploying virtual machines across multiple resource groups provides no protection against datacenter-level failures, as resources in different resource groups can still reside in the same physical datacenter.

Overall explanation: Availability zones are unique physical locations within an Azure region, with each zone consisting of one or more datacenters equipped with independent infrastructure. By distributing virtual machines across availability zones, organizations can protect their applications from datacenter-level failures while maintaining low latency and high performance within a single region. This configuration enables the 99.99% availability SLA for virtual machines that Microsoft offers when deployed across two or more availability zones.

Question 18

A government agency in the United States needs to deploy Azure services to host applications that process sensitive federal data. The agency must comply with specific regulations that require physical and logical isolation of data within the United States, screened US personnel for operations, and compliance with FedRAMP High standards. Which Azure deployment option should the agency select?

-  **Deploy resources in Azure China regions operated by 21Vianet**
 - This answer is incorrect. Azure China regions operated by 21Vianet is a sovereign cloud designed to meet Chinese regulatory requirements for data residency and compliance within China. This deployment option would not meet the US government's requirements and would actually violate data sovereignty needs by placing US federal data outside United States jurisdiction.
-  **Deploy resources in Azure public cloud regions located in the United States (such as East US or West US)**
 - This answer is incorrect. While Azure public cloud regions in the United States provide geographic data residency, they do not offer the physical and logical isolation, screened personnel, or dedicated FedRAMP High compliance frameworks required for sensitive US federal government data. Public regions serve general commercial customers and do not meet specialized government compliance requirements.
-  **Deploy resources using Azure Arc on the agency's on-premises infrastructure**
 - This answer is incorrect. Azure Arc extends Azure management capabilities to on-premises and multi-cloud environments but does not provide the sovereign cloud benefits needed for this scenario. While the agency could use on-premises infrastructure, the question specifically asks for an Azure deployment option that meets government compliance requirements, which Azure Government provides as a managed sovereign cloud solution.
-  **Deploy resources in Azure Government**
 - This answer is correct. Azure Government is a sovereign cloud designed specifically for US government agencies and their partners, providing physical and logical isolation from Azure public cloud. It features screened US personnel, dedicated datacenters on US soil, and built-in compliance with FedRAMP High

and other government-specific regulatory requirements, making it the appropriate choice for this scenario.

Overall explanation: Sovereign regions like Azure Government and Azure China are physically and logically separated instances of Azure designed to meet specific compliance, regulatory, and data residency requirements for government entities and regulated industries. Azure Government specifically addresses US federal, state, and local government needs with dedicated infrastructure, screened personnel, and specialized compliance certifications that standard public cloud regions do not provide.

Question 19

Your company is planning to migrate an on-premises application server to Azure. The application requires 8 vCPUs, 32 GB of RAM, and runs on Windows Server 2022. You need to recommend an Azure solution that provides full control over the operating system, allows installation of custom software, and supports the required specifications. What should you recommend?

-  **Azure Container Instances**
 - This answer is incorrect. Azure Container Instances is designed for running containerized applications and does not provide full control over the operating system in the traditional sense. While containers can run Windows Server containers, they operate with a shared kernel model and don't offer the same level of OS control as virtual machines, making it unsuitable for this scenario.
-  **Azure App Service**
 - This answer is incorrect. Azure App Service is a Platform as a Service (PaaS) offering designed for hosting web applications, REST APIs, and mobile backends. It does not provide full control over the operating system or allow direct installation of custom software at the OS level, which are requirements specified in the scenario.
-  **Azure Virtual Machines**
 - This answer is correct. Azure Virtual Machines is an Infrastructure as a Service (IaaS) offering that provides complete control over the operating system, supports custom software installation, and allows you to select VM sizes with specific vCPU and memory configurations. This solution meets all the

requirements including the ability to run Windows Server 2022 with the specified 8 vCPUs and 32 GB RAM.

- **✗ Azure Functions**

- This answer is incorrect. Azure Functions is a serverless compute service designed for running event-driven code without managing infrastructure. It does not provide control over the operating system, does not allow custom software installation at the OS level, and is not designed for migrating traditional application servers that require specific OS configurations.

Overall explanation: Azure Virtual Machines is the appropriate solution when you need full control over the operating system, require specific hardware configurations, and need to install custom software. As an IaaS offering, VMs provide the flexibility to configure the environment exactly as needed, making them ideal for lift-and-shift migrations from on-premises servers to Azure while maintaining complete administrative control.

Question 20

Your company has been running multiple Azure resources for six months, including virtual machines, storage accounts, and databases. The IT manager asks you to identify opportunities to reduce costs, improve security, and optimize performance across all Azure resources. Which Azure service should you use to get personalized recommendations based on your actual resource usage and configuration?

- **✗ Microsoft Defender for Cloud**

- This answer is incorrect. Microsoft Defender for Cloud is primarily focused on security posture management and threat protection for cloud workloads. While it does provide security recommendations, it does not offer comprehensive guidance on cost optimization, performance improvements, or operational excellence like the service described in the question.

- **✗ Azure Monitor**

- This answer is incorrect. Azure Monitor is used for collecting, analyzing, and acting on telemetry data from Azure and on-premises environments. While it helps you understand how applications are performing and identify issues, it does not provide personalized recommendations for cost optimization, security, reliability, operational excellence, or performance improvements.

-  **Azure Service Health**
 - This answer is incorrect. Azure Service Health provides personalized alerts and guidance when Azure service issues affect your resources. It helps you monitor the health of Azure services and your resources during planned maintenance and outages, but it does not analyze your configurations to provide optimization recommendations.
-  **Azure Advisor**
 - This answer is correct. Azure Advisor is a personalized cloud consultant that analyzes your resource configuration and usage telemetry to provide recommendations across five categories: Cost, Security, Reliability, Operational Excellence, and Performance. It provides actionable recommendations specific to your actual Azure deployment, making it the ideal tool for this scenario.

Overall explanation: Azure Advisor serves as a free, personalized cloud consultant that continuously analyzes your Azure resources and provides best-practice recommendations across five key pillars: Cost, Security, Reliability, Operational Excellence, and Performance. It is specifically designed to help organizations optimize their Azure deployments by providing actionable, prioritized recommendations based on actual resource usage patterns and configurations.

Question 21

Your company runs multiple virtual machines in Azure that host a critical web application. The IT manager wants to be automatically notified via email whenever the CPU usage on any of these virtual machines exceeds 85% for more than 5 minutes. Which Azure Monitor capability should you configure to meet this requirement?

-  **Azure Monitor metrics chart**
 - This answer is incorrect. Azure Monitor metrics charts are visualization tools used to display performance data and trends in graphical format within the Azure portal. While they help you view CPU usage and other metrics, they do not provide automated notification capabilities or the ability to trigger actions based on threshold conditions.
-  **Log Analytics workspace query**

- This answer is incorrect. While Log Analytics workspaces can be used to query and analyze monitoring data collected from Azure resources, they do not automatically send notifications based on specific conditions. Log Analytics is primarily used for storing and querying log data, not for generating automated alerts based on performance thresholds.
-  **Application Insights availability test**
 - This answer is incorrect. Application Insights availability tests are designed to monitor the availability and responsiveness of web applications by sending requests from multiple geographic locations. These tests focus on application endpoint availability rather than infrastructure metrics like CPU usage, and they would not be appropriate for monitoring virtual machine performance thresholds.
-  **Azure Monitor alert rule**
 - This answer is correct. Azure Monitor alert rules allow you to define specific conditions (such as CPU usage exceeding 85%) and configure automated responses including email notifications when those conditions are met. Alert rules can monitor metrics, evaluate them against thresholds over specified time periods, and trigger action groups to send notifications to designated recipients.

Overall explanation: Azure Monitor alert rules are the correct solution for automatically monitoring resource metrics and sending notifications when specified conditions are met. They combine metric evaluation, threshold comparison, time-based conditions, and action groups (which can include email notifications) into a single automated monitoring solution. This scenario represents a common use case for Azure Monitor alerts in production environments.

Question 22

Your organization's IT manager asks you to demonstrate how to quickly find and deploy an Azure Virtual Machine without using command-line tools or scripts. The manager wants to use a graphical interface that works from any web browser without installing additional software. Which Azure management tool should you demonstrate?

-  **Azure CLI**
 - This answer is incorrect. Azure CLI (Command-Line Interface) is a command-line tool that uses text-based commands rather than a graphical interface. Although it can be accessed through Azure Cloud Shell in a browser, it still requires typing

commands and does not provide the graphical, point-and-click experience requested by the manager.

- **Azure Mobile App**

- This answer is incorrect. While the Azure Mobile App does provide a graphical interface and works without command-line tools, it is designed for monitoring and basic management tasks on mobile devices rather than comprehensive resource deployment. It has limited functionality compared to the full Azure Portal and is not the optimal tool for demonstrating VM deployment workflows.

- **Azure PowerShell**

- This answer is incorrect. Azure PowerShell is a command-line tool that requires installation and uses script-based commands rather than a graphical interface. While PowerShell is powerful for automation and management tasks, it does not meet the requirement for a graphical interface and requires users to learn PowerShell cmdlets and syntax.

- **Azure Portal**

- This answer is correct. The Azure Portal is a web-based, graphical user interface accessible from any web browser without requiring additional software installation. It provides a comprehensive GUI for managing Azure resources, including creating and deploying virtual machines through an intuitive, step-by-step interface that meets all the manager's requirements.

Overall explanation: The Azure Portal is Microsoft's primary web-based management interface, designed specifically to provide a comprehensive graphical experience for managing all Azure services without requiring any software installation beyond a web browser. It offers wizards, forms, and visual tools that make it ideal for users who prefer GUI-based management over command-line interfaces, making it the perfect solution for this scenario.

Question 23

Your company is deploying a business-critical application in Azure that requires high availability and disaster recovery capabilities. The application must remain available even if an entire Azure region becomes unavailable due to a natural disaster. You need to ensure that Azure automatically replicates certain platform services to a secondary location for disaster recovery purposes. Which Azure feature should you rely on to achieve this geographic redundancy?

-  **Azure region pairs**
 - This answer is correct. Azure region pairs are two regions within the same geography that are paired together for disaster recovery purposes. Azure automatically replicates certain platform services and updates to the paired region, ensuring that if one region experiences an outage, services can fail over to the paired region, providing geographic redundancy and business continuity.
-  **Availability Sets in multiple regions**
 - This answer is incorrect. Availability Sets are used to protect applications from hardware failures within a single datacenter by distributing virtual machines across fault domains and update domains. They operate within a single region and datacenter, and do not provide cross-region redundancy or automatic replication to secondary locations.
-  **Availability Zones within a single region**
 - This answer is incorrect. Availability Zones provide high availability within a single Azure region by distributing resources across physically separate datacenters, but they do not protect against a region-wide outage. Availability Zones are designed for redundancy within the same geographic region, not across different regions.
-  **Resource groups spanning multiple regions**
 - This answer is incorrect. Resource groups are logical containers for Azure resources and do not inherently provide any replication or disaster recovery capabilities. While you can deploy resources to multiple regions and organize them in resource groups, resource groups themselves do not automatically replicate data or provide geographic redundancy.

Overall explanation: Azure region pairs are specifically designed to provide disaster recovery and business continuity across geographic locations. Each Azure region is paired with another region within the same geography (at least 300 miles apart when possible), and Azure uses these pairs to ensure platform resilience by sequencing updates and providing automatic replication for certain services like geo-redundant storage. This is the fundamental Azure feature for protecting against region-wide failures.

Question 24

Your organization wants to ensure that users accessing cloud applications from outside the corporate network must perform multi-factor authentication (MFA), while users on the corporate network can sign in with just their username and password. Which Microsoft Entra feature should you implement to accomplish this requirement?

-  **Microsoft Entra Conditional Access**
 - This answer is correct. Microsoft Entra Conditional Access allows administrators to create policies that enforce specific access controls based on conditions such as user location, device state, application, or risk level. In this scenario, you would create a Conditional Access policy that requires MFA as a grant control when the sign-in location is outside the corporate network, while allowing standard authentication from trusted locations.
-  **Microsoft Entra Privileged Identity Management**
 - This answer is incorrect. Microsoft Entra Privileged Identity Management (PIM) is used to manage, control, and monitor access to privileged roles and resources within an organization. It provides just-in-time access and approval workflows for administrative roles, but it does not create location-based authentication policies for general users accessing applications.
-  **Microsoft Entra ID Protection**
 - This answer is incorrect. Microsoft Entra ID Protection is designed to detect and respond to identity-based risks such as unusual sign-in patterns, compromised credentials, and risky user behaviors. While it can trigger risk-based policies, it is not the primary tool for creating location-based access policies that conditionally require MFA.
-  **Microsoft Entra Multi-Factor Authentication**
 - This answer is incorrect. While Microsoft Entra Multi-Factor Authentication is the technology that provides the additional authentication factor, it does not by itself provide the conditional logic to determine when MFA should be required. MFA must be enforced through a policy mechanism like Conditional Access to apply it selectively based on conditions such as network location.

Overall explanation: Microsoft Entra Conditional Access is the appropriate solution for implementing policy-based access controls that respond to specific signals or conditions. It

evaluates conditions such as user identity, location, device compliance, and application sensitivity to determine whether to grant access, require additional authentication, or block access entirely. This makes it ideal for scenarios requiring different authentication requirements based on where users are connecting from.

Question 25

A company wants to provide its 500 remote employees with access to Windows 11 desktops and Microsoft 365 applications. The IT department needs to centrally manage these desktops, apply security policies consistently, and minimize the hardware requirements on employee devices. Which Azure solution should the company implement?

-  **Microsoft 365 Business Premium with local desktop installations**
 - This answer is incorrect. Microsoft 365 Business Premium provides productivity applications and security features but requires local installations on physical devices and does not provide virtualized desktop infrastructure. This solution would not minimize hardware requirements on employee devices or provide the centralized desktop management that Azure Virtual Desktop offers.
-  **Azure Virtual Desktop**
 - This answer is correct. Azure Virtual Desktop is specifically designed to provide virtualized Windows desktops and applications to remote users with centralized management, security, and minimal client-side hardware requirements. It supports Windows 11 multi-session capabilities, integrates seamlessly with Microsoft 365, and allows IT departments to manage desktops centrally while users can access them from virtually any device.
-  **Azure App Service with web-based application hosting**
 - This answer is incorrect. Azure App Service is a platform-as-a-service (PaaS) offering designed for hosting web applications, APIs, and mobile backends, not for providing full Windows desktop experiences. It cannot deliver the Windows 11 desktop environment and centralized desktop management capabilities that the scenario requires.
-  **Azure Virtual Machines running Windows 11 with Remote Desktop Protocol (RDP) access for each user**

- This answer is incorrect. While Azure Virtual Machines could technically provide Windows 11 access via RDP, this approach would require provisioning and managing individual VMs for each user, resulting in higher costs and administrative overhead. Azure Virtual Desktop is specifically designed for this desktop virtualization scenario and provides multi-session capabilities that make it more efficient and cost-effective for multiple users.

Overall explanation: Azure Virtual Desktop is Microsoft's desktop and application virtualization service that runs in Azure, specifically designed for scenarios requiring remote access to Windows desktops and applications. It provides cost-effective multi-session Windows 11 capabilities, centralized management, enhanced security through Microsoft Entra ID integration, and allows users to access their desktops from various devices with minimal local hardware requirements, making it the ideal solution for this scenario.

Question 26

Your company's development team currently deploys Azure resources by manually clicking through the Azure portal. The operations manager wants to improve consistency, reduce deployment errors, and enable the team to version control their infrastructure configurations. Which approach should the team implement?

- **✗ Schedule regular backups of all Azure resources using Azure Backup**
 - This answer is incorrect. Azure Backup is designed for data protection and disaster recovery, not for managing infrastructure deployment consistency or version control. Backups do not prevent deployment errors or improve the deployment process itself; they only provide recovery options after issues occur.
- **✗ Assign multiple administrators to review each manual deployment before execution**
 - This answer is incorrect. Adding more reviewers to manual deployments may catch some errors but does not address the fundamental issues of consistency or version control. This approach actually increases the time required for deployments and still relies on error-prone manual processes through the portal.
- **✗ Create detailed documentation with step-by-step screenshots of the portal deployment process**
 - This answer is incorrect. While documentation helps standardize processes, it does not eliminate manual deployment through the portal, which remains

error-prone and doesn't provide version control for the infrastructure itself. Documentation can become outdated and still requires manual interpretation and execution by team members.

-  **Use Azure Resource Manager (ARM) templates to define and deploy infrastructure**
 - This answer is correct. Azure Resource Manager (ARM) templates are a form of Infrastructure as Code (IaC) that allow teams to define infrastructure in JSON files, which can be version controlled, tested, and deployed consistently. This directly addresses all the requirements: improving consistency, reducing errors, and enabling version control of infrastructure configurations.

Overall explanation: Infrastructure as Code (IaC) is the practice of managing and provisioning infrastructure through machine-readable definition files rather than manual processes. ARM templates, along with other tools like Bicep or Terraform, enable teams to treat infrastructure the same way as application code—with version control, testing, and automated deployment. This approach directly solves the problems of inconsistency and errors inherent in manual, portal-based deployments.

Question 27

Your organization has a resource group named RG-Production that contains several virtual machines. You need to ensure that a user named User1 can start and stop the virtual machines in RG-Production but cannot delete the virtual machines or modify their configurations. What should you do?

-  **Create and assign User1 a custom Azure RBAC role scoped to RG-Production that allows only the required VM start and stop actions**
 - This answer is correct. Creating a custom Azure RBAC role allows you to grant only the specific permissions required—such as starting and stopping virtual machines—while excluding permissions to delete or modify them.
-  **Assign User1 the Virtual Machine Contributor role on RG-Production.**
 - This answer is incorrect. The Virtual Machine Contributor role is a built-in Azure RBAC role that allows full management of virtual machines, including starting, stopping, restarting, modifying, and deleting VMs. This is because it includes permissions such as Microsoft.Compute/virtualMachines/*.

- **✗ Assign User1 the Contributor role on RG-Production.**
 - This answer is incorrect. While the Contributor role would allow User1 to start and stop virtual machines, it grants too many permissions for the stated requirement. The Contributor role allows users to create, modify, and delete resources, which exceeds the requirement since User1 should not be able to delete virtual machines or modify their configurations.
- **✗ Assign User1 the Reader role on RG-Production.**
 - This answer is incorrect. The Reader role provides read-only access to Azure resources, allowing users to view resources but not make any changes. User1 would not be able to start or stop the virtual machines with this role, as these are considered management operations that modify the state of the resource.

Overall explanation: Azure RBAC uses built-in roles to grant specific permissions at different scopes (management group, subscription, resource group, or resource level). The Virtual Machine Contributor role is designed specifically to provide virtual machine management capabilities without granting excessive permissions, following the principle of least privilege. This scenario demonstrates how to select the appropriate built-in role that matches the required permissions without over-provisioning access rights.

Question 28

A company is migrating their on-premises infrastructure to Azure. They have a development environment that is only used during business hours (9 AM to 5 PM, Monday through Friday) and a production environment that must run continuously. The IT manager wants to minimize costs while maintaining the required availability. Which pricing model should they implement for the development environment to achieve the greatest cost savings?

- **✗ Azure Hybrid Benefit to use existing on-premises licenses for the development virtual machines**
 - This answer is incorrect. While Azure Hybrid Benefit does reduce costs by allowing customers to use existing Windows Server and SQL Server licenses on Azure, it provides savings on licensing costs rather than addressing the usage pattern issue. The company would still pay for compute resources running 24/7, which is unnecessary for a development environment used only during business hours.

-  **Pay-as-you-go pricing with automated shutdown and startup schedules for the development virtual machines**
 - This answer is correct. Pay-as-you-go pricing with automated shutdown schedules allows the company to pay only for the compute hours actually used (40 hours per week instead of 168 hours). By automatically stopping the development VMs outside business hours and starting them when needed, the company eliminates charges for approximately 76% of the week, resulting in the greatest cost savings for this intermittent usage pattern.
-  **Reserved Instances with a one-year commitment for the development virtual machines**
 - This answer is incorrect. Reserved Instances provide cost savings of up to 72% compared to pay-as-you-go pricing, but they require payment for 24/7 availability regardless of whether the resources are running. Since the development environment only needs to run 40 hours per week (approximately 24% of total hours), paying for continuous reservation would waste money on unused capacity.
-  **Reserved Instances with a three-year commitment for the development virtual machines**
 - This answer is incorrect. A three-year Reserved Instance commitment provides the deepest discount on reserved capacity (compared to one-year reservations), but like the one-year option, it requires payment for continuous availability. This pricing model is most suitable for workloads with consistent, predictable usage patterns like the production environment, not for intermittent development workloads.

Overall explanation: Cloud pricing models must align with actual usage patterns to maximize cost efficiency. For workloads with intermittent or predictable schedules, pay-as-you-go pricing combined with resource scheduling provides optimal savings by charging only for consumed resources. Reserved Instances are better suited for steady-state workloads that require 24/7 availability, while pay-as-you-go offers flexibility for variable usage patterns.

Question 29

A software development company needs to deploy a web application that experiences unpredictable traffic spikes throughout the day. The application consists of multiple microservices that need to scale independently based on demand. The company wants to

minimize infrastructure management overhead while maintaining the ability to package applications with their dependencies. Which Azure compute option best meets these requirements?

-  **Azure Container Instances with individual container deployment**
 - This answer is incorrect. Azure Container Instances (ACI) is suitable for simple, isolated container workloads but lacks the orchestration capabilities needed for managing multiple interdependent microservices. ACI does not provide built-in features for service discovery, load balancing between microservices, or sophisticated auto-scaling policies that would be essential for a complex microservices application.
-  **Azure Kubernetes Service (AKS) with auto-scaling enabled**
 - This answer is correct. Azure Kubernetes Service (AKS) is specifically designed for orchestrating containerized microservices applications with features like automatic scaling, self-healing, and service discovery. AKS minimizes infrastructure management as a managed service while allowing each microservice to scale independently based on its specific resource demands, perfectly matching the requirement for unpredictable traffic patterns and minimal overhead.
-  **Azure App Service with a single web app deployment**
 - This answer is incorrect. While Azure App Service offers auto-scaling and reduced management overhead, it is optimized for monolithic web applications rather than microservices architectures. Deploying multiple microservices as a single web app would prevent independent scaling of individual services, which is a key requirement in this scenario, and would not leverage the benefits of containerized deployment with dependencies.
-  **Azure Virtual Machines with manual scaling configuration**
 - This answer is incorrect. While Azure Virtual Machines provide full control over the infrastructure, they require significant management overhead for scaling, patching, and maintenance. Manual scaling configuration would not efficiently handle unpredictable traffic spikes, and VMs do not inherently support the microservices architecture pattern or independent scaling of individual services.

Overall explanation: This question tests understanding of different Azure compute options and their appropriate use cases, particularly focusing on containers versus virtual machines and their scaling capabilities. AKS is the optimal choice because it combines containerization

benefits (packaging with dependencies), orchestration for microservices, managed infrastructure (low overhead), and sophisticated auto-scaling capabilities that support independent scaling of services—all critical requirements in the scenario.

Question 30

A manufacturing company currently maintains all its servers and infrastructure in an on-premises datacenter. The company wants to migrate some workloads to Azure but must keep its proprietary production control systems on-premises due to regulatory requirements and the need for ultra-low latency with factory floor equipment. The IT director wants to use a consistent set of management tools across both environments and enable seamless connectivity between on-premises and cloud resources. Which cloud model should the company implement?

-  **Community cloud**
 - This answer is incorrect. A community cloud is a cloud infrastructure shared by several organizations with common concerns (such as security, compliance, or jurisdiction), typically managed by the organizations or a third party. This model does not address the specific requirement of integrating on-premises infrastructure with public cloud services, which is essential for this manufacturing company's scenario.
-  **Public cloud**
 - This answer is incorrect. A public cloud model involves using only cloud-based resources provided by a cloud service provider like Microsoft Azure, with no integration of on-premises infrastructure. This approach would not allow the company to keep its production control systems on-premises while also utilizing Azure services, which is a key requirement in this scenario.
-  **Private cloud**
 - This answer is incorrect. A private cloud is a cloud environment dedicated exclusively to one organization, typically hosted on-premises or by a third party, but it does not involve integration with public cloud services. This model would not provide the company with access to Azure's cloud services while maintaining their on-premises systems, nor would it offer the unified management across both environments that the company needs.
-  **Hybrid cloud**

- This answer is correct. A hybrid cloud model combines on-premises infrastructure with public cloud services, allowing the company to keep sensitive production control systems on-premises while migrating other workloads to Azure. This model enables seamless connectivity between environments, supports consistent management tools across both platforms, and meets both the regulatory requirements and the low-latency needs for factory equipment.

Overall explanation: The hybrid cloud model is specifically designed for scenarios where organizations need to maintain some resources on-premises while leveraging public cloud services for other workloads. This approach provides the flexibility to meet regulatory requirements, performance needs, and legacy system constraints while still benefiting from cloud scalability and services. Azure offers numerous hybrid cloud solutions, including Azure Arc and Azure Stack, to enable unified management and seamless integration between on-premises and cloud environments.

Question 31

Your company is deploying a web application on Azure and the security team wants to implement the defense-in-depth model. The application uses Azure Virtual Machines, Azure SQL Database, and stores sensitive customer data. Which approach best demonstrates the implementation of defense-in-depth for this scenario?

- **✗ Implement only network security groups (NSGs) with strict rules to control all traffic to and from the virtual machines and database.**
 - This answer is incorrect. While network security groups are an important security control, relying on a single layer of security violates the defense-in-depth principle. Defense-in-depth requires implementing multiple independent layers of security controls so that if one layer fails, other layers continue to provide protection.
- **✗ Configure role-based access control (RBAC) for all resources and implement Microsoft Entra ID authentication for application users.**
 - This answer is incorrect. While RBAC and Microsoft Entra ID authentication are important security controls for identity and access management, they represent only one layer of security. Defense-in-depth requires implementing security controls across multiple layers including physical, network, compute, application, and data layers to provide comprehensive protection.

-  **Implement multiple layers of security including network security groups, encryption at rest for the database, multi-factor authentication with Microsoft Entra ID, and Microsoft Defender for Cloud monitoring.**
 - This answer is correct. This option demonstrates true defense-in-depth by implementing security controls across multiple layers: network security (NSGs), data security (encryption at rest), identity and access (multi-factor authentication with Microsoft Entra ID), and monitoring (Microsoft Defender for Cloud). This multi-layered approach ensures that if one security control fails or is breached, additional layers provide continued protection.
-  **Focus security efforts exclusively on the perimeter by implementing a firewall and ensuring all virtual machines are updated with the latest security patches.**
 - This answer is incorrect. Focusing exclusively on perimeter security represents a single-layer security approach, which does not align with the defense-in-depth model. Defense-in-depth requires security controls at multiple layers throughout the architecture, not just at the perimeter, to protect against various attack vectors and potential breaches.

Overall explanation: The defense-in-depth model is a security strategy that implements multiple layers of security controls throughout an IT system. The principle ensures that if one defensive layer is compromised, additional layers continue to provide protection. The correct option demonstrates this by combining network controls, data encryption, identity management, and monitoring across different layers of the architecture.

Question 32

Your organization needs to deploy identical development, testing, and production environments in Azure. Each environment must contain the same virtual machines, storage accounts, and virtual networks with identical configurations. The IT manager wants to ensure consistency across all deployments and minimize manual configuration errors. Which Azure feature should you recommend?

-  **Azure Resource Manager (ARM) templates**
 - This answer is correct. ARM templates are JSON files that define the infrastructure and configuration for Azure resources using declarative syntax. They enable you to deploy multiple identical environments consistently and repeatedly, ensuring all resources are configured the same way across

development, testing, and production environments while minimizing human error.

- **✗ Azure Portal dashboard**

- This answer is incorrect. Azure Portal dashboards provide a customizable view for monitoring and managing Azure resources through visualizations and shortcuts. Dashboards are useful for visualization and management but do not provide any capability to deploy or configure infrastructure resources or ensure consistency across environments.

- **✗ Resource groups**

- This answer is incorrect. Resource groups are logical containers that hold related Azure resources for management purposes. While you would use resource groups to organize resources in each environment, they do not provide the capability to define, automate, or replicate infrastructure deployments consistently across multiple environments.

- **✗ Azure Policy**

- This answer is incorrect. Azure Policy is used to enforce organizational standards and assess compliance at scale, such as ensuring resources are deployed only in certain regions or have specific tags. While Azure Policy helps maintain governance, it does not deploy or configure the actual infrastructure resources needed for the environments.

Overall explanation: ARM templates are the appropriate solution for deploying consistent, repeatable infrastructure across multiple environments. They use infrastructure as code (IaC) principles to define all resources declaratively, which can be version-controlled, tested, and reused to create identical environments. This approach significantly reduces manual configuration errors and ensures consistency across development, testing, and production deployments.

Question 33

Your company is migrating a web application to Azure. The application will run on Azure Virtual Machines in an Infrastructure as a Service (IaaS) deployment model. The IT manager asks you to identify which security tasks will remain the company's responsibility after the migration. Which of the following is the company responsible for managing?

-  **Security patches and updates for the operating system running on the virtual machines**
 - This answer is correct. In an IaaS model with Azure Virtual Machines, the customer is responsible for managing the guest operating system, including installing security patches and updates. This is a key distinction of the shared responsibility model where Microsoft manages the infrastructure, but customers manage everything from the OS layer upward, including applications, data, and OS-level security.
-  **Physical security of the datacenter hosting the virtual machines**
 - This answer is incorrect. Physical security of the datacenter is always Microsoft's responsibility across all cloud service models (IaaS, PaaS, and SaaS). Microsoft manages the physical facilities, including access controls, surveillance, and environmental protections for all Azure datacenters.
-  **Security of the Azure hypervisor layer**
 - This answer is incorrect. The hypervisor layer is part of the virtualization infrastructure that Microsoft manages and maintains. Microsoft is responsible for securing the hypervisor to ensure proper isolation between customer virtual machines and the integrity of the virtualization platform.
-  **Physical network infrastructure connecting the Azure datacenter**
 - This answer is incorrect. The physical network infrastructure, including routers, switches, and cabling within Azure datacenters, is Microsoft's responsibility. Microsoft manages all physical infrastructure components to ensure connectivity and availability of Azure services.

Overall explanation: The shared responsibility model defines which security and management tasks belong to Microsoft versus the customer, and this varies by service model. In IaaS deployments like Azure Virtual Machines, while Microsoft handles physical infrastructure, networking hardware, and the virtualization layer, customers assume responsibility for the operating system and everything above it, including OS patches, applications, data, and access management. Understanding this division of responsibilities is fundamental to properly securing Azure workloads.

Question 34

A company is planning to migrate their on-premises infrastructure to Azure. They need to estimate the monthly cost of running 5 virtual machines, 2 Azure SQL databases, and storing 10 TB of data in Azure Blob Storage across two different Azure regions. Which Azure tool should the IT manager use to calculate these estimated costs before deploying any resources?

-  **Total Cost of Ownership (TCO) Calculator**
 - This answer is incorrect. The Total Cost of Ownership (TCO) Calculator is designed to compare the cost of running workloads on-premises versus in Azure over a specified time period, helping justify cloud migration decisions. While useful for migration planning, it does not provide detailed pricing estimates for specific Azure services and configurations like the scenario requires.
-  **Azure Advisor**
 - This answer is incorrect. Azure Advisor is a recommendation engine that analyzes your existing deployed Azure resources and provides best practice recommendations for optimization, security, reliability, performance, and cost reduction. It cannot estimate costs for resources that have not yet been deployed to Azure.
-  **Azure Pricing Calculator**
 - This answer is correct. The Azure Pricing Calculator is specifically designed to estimate costs for Azure services before deployment by allowing users to select and configure various Azure products, specify configurations like region and tier, and receive detailed monthly cost estimates. This tool is ideal for planning and budgeting migration scenarios where no resources have been deployed yet.
-  **Azure Cost Management and Billing**
 - This answer is incorrect. Azure Cost Management and Billing is used to monitor, analyze, and optimize costs for resources that are already deployed and running in Azure. While it provides valuable insights into actual spending and cost trends, it cannot be used to estimate costs for resources that don't exist yet in your Azure environment.

Overall explanation: The Azure Pricing Calculator is the appropriate tool for estimating costs of Azure services before deployment, allowing users to configure various services with specific parameters and receive detailed cost breakdowns. This pre-deployment estimation capability

differentiates it from tools like Azure Cost Management (which tracks actual spending) and Azure Advisor (which optimizes existing resources). The TCO Calculator serves a different purpose by comparing on-premises versus cloud costs rather than providing granular Azure service pricing.

Question 35

Your organization needs to deploy 15 identical virtual machines across three different Azure regions for a new application. The IT manager wants to ensure consistency in the configuration of all virtual machines and simplify the deployment process. Which Azure Resource Manager capability would be most appropriate for this scenario?

-  **Use Azure Resource Manager resource groups to organize the virtual machines by region**
 - This answer is incorrect. Although resource groups are important for organizing and managing related Azure resources, they are containers for resources and do not inherently ensure consistency in configuration or simplify the deployment of identical resources. Resource groups alone do not provide the automation or template-based deployment needed for this scenario.
-  **Use Azure Resource Manager locks to prevent changes to the virtual machines after deployment**
 - This answer is incorrect. While Azure Resource Manager locks are useful for preventing accidental deletion or modification of resources, they do not help with the initial deployment or ensure consistency across multiple resources. Locks are a governance feature, not a deployment tool, and would not address the requirement to simplify deploying 15 identical virtual machines.
-  **Use Azure Resource Manager role-based access control (RBAC) to manage permissions for the virtual machines**
 - This answer is incorrect. Azure Resource Manager RBAC is used to control who can access and manage Azure resources by assigning permissions, but it does not help with deploying infrastructure or ensuring configuration consistency. RBAC addresses security and access management, not the deployment requirements described in the scenario.
-  **Use Azure Resource Manager templates to define the infrastructure and deploy it consistently across all regions**

- This answer is correct. Azure Resource Manager templates (ARM templates or Bicep) allow you to define infrastructure as code, enabling you to deploy identical resources consistently across multiple regions. This approach ensures all 15 virtual machines have the same configuration and significantly simplifies the deployment process through automation and repeatability.

Overall explanation: Azure Resource Manager templates provide infrastructure-as-code capabilities that allow organizations to define, deploy, and manage Azure resources consistently and repeatedly. For scenarios requiring multiple identical deployments across regions, ARM templates are the most appropriate solution as they ensure configuration consistency, enable automation, and simplify management of complex deployments.

Question 36

A development team needs to deploy a small application that processes incoming HTTP requests and performs simple data transformations. The application runs for only a few seconds per request and receives sporadic traffic throughout the day. The team wants to minimize infrastructure management and only pay for actual compute time used. Which Azure compute service should they recommend?

- **Azure Functions**

- This answer is correct. Azure Functions is a serverless compute service that automatically scales based on demand and charges only for the time code is executing. It is ideal for event-driven workloads like processing HTTP requests with sporadic traffic, requires minimal infrastructure management, and follows a consumption-based pricing model.

- **Azure App Service**

- This answer is incorrect. While Azure App Service can host HTTP-triggered applications and offers Platform-as-a-Service (PaaS) benefits, it typically requires an App Service Plan that runs continuously and incurs charges even during idle periods. This does not meet the requirement to pay only for actual compute time used for sporadic workloads.

- **Azure Virtual Machines**

- This answer is incorrect. Azure Virtual Machines require ongoing management of the underlying infrastructure, including OS updates and patching. Additionally,

VMs incur costs whenever they are running, even during idle periods, which does not align with the requirement to pay only for actual compute time used.

- **Azure Container Instances**

- This answer is incorrect. Azure Container Instances provides on-demand containers without orchestration overhead, but it charges for the entire time the container is allocated, not just execution time. For sporadic, short-duration HTTP requests, this would result in higher costs compared to a true consumption-based serverless option.

Overall explanation: Azure Functions is specifically designed for event-driven, serverless compute scenarios where code runs in response to triggers like HTTP requests. The consumption-based pricing model charges only for actual execution time (measured in gigabyte-seconds), making it the most cost-effective solution for sporadic workloads with short execution times. This eliminates infrastructure management while optimizing costs for unpredictable traffic patterns.

Question 37

Your organization has recently migrated several workloads to Azure, including virtual machines, storage accounts, and Azure SQL databases. The IT security team needs to continuously assess the security posture of these resources, receive recommendations to improve security, and be alerted about potential threats. Which Azure service should they implement to meet these requirements?

- **Microsoft Defender for Cloud**

- This answer is correct. Microsoft Defender for Cloud is designed to continuously assess the security posture of Azure resources, provide security recommendations based on best practices, and detect threats across hybrid cloud environments. It provides a secure score, actionable recommendations to improve security configurations, and alerts for potential security threats, which directly addresses all the requirements specified in the scenario.

- **Azure Policy**

- This answer is incorrect. Azure Policy is used to enforce organizational standards and assess compliance at scale by creating, assigning, and managing policies that control resource properties. While it helps maintain governance and compliance, it does not provide security posture assessment, threat detection, or

security-specific recommendations in the comprehensive manner that the scenario requires.

- **✗ Microsoft Entra ID Protection**

- This answer is incorrect. Microsoft Entra ID Protection is specifically designed to detect and remediate identity-based risks and protect user accounts from compromise. While it provides security for identity resources, it does not assess the security posture of Azure infrastructure resources like virtual machines, storage accounts, or databases, nor does it provide security recommendations for these resources.

- **✗ Azure Monitor**

- This answer is incorrect. Azure Monitor is a comprehensive monitoring solution that collects, analyzes, and acts on telemetry data from Azure and on-premises environments to maximize performance and availability. While it can alert on operational metrics and logs, it is not specifically designed for security posture assessment, security recommendations, or threat detection across Azure resources.

Overall explanation: Microsoft Defender for Cloud is Azure's cloud-native application protection platform (CNAPP) that provides unified security management and threat protection across hybrid cloud workloads. It specifically addresses the three key requirements in the scenario: continuous security posture assessment through secure scores, actionable security recommendations to harden resources, and advanced threat protection with security alerts. This makes it the primary service for comprehensive cloud security management in Azure.

Question 38

Contoso Ltd. currently runs several business applications on physical servers in their on-premises datacenter. The company wants to modernize their infrastructure by migrating some applications to Azure while keeping other applications on-premises due to regulatory requirements. Contoso also needs to ensure that users can authenticate seamlessly across both environments. Which cloud deployment model does this scenario describe?

- **✗ Private cloud**

- This answer is incorrect. A private cloud refers to cloud computing resources used exclusively by a single organization, typically hosted on-premises or in a dedicated environment. While Contoso maintains on-premises servers, they are

also utilizing Azure's public cloud services, which means this is not solely a private cloud deployment.

- **Hybrid cloud**

- This answer is correct. A hybrid cloud combines on-premises infrastructure (private cloud) with public cloud services (Azure), allowing data and applications to be shared between them. Contoso's scenario perfectly demonstrates hybrid cloud by running some applications on-premises while migrating others to Azure, with seamless authentication across both environments using services like Microsoft Entra Connect.

- **Public cloud**

- This answer is incorrect. A public cloud deployment model means that all resources and applications run on cloud service provider infrastructure (like Azure) and are accessed over the internet. In this scenario, Contoso is maintaining on-premises infrastructure alongside Azure resources, which goes beyond a purely public cloud model.

- **Community cloud**

- This answer is incorrect. A community cloud is a deployment model where infrastructure is shared by several organizations with common concerns (such as security, compliance, or jurisdiction). Contoso's scenario describes a single organization using both on-premises and Azure resources, not multiple organizations sharing cloud infrastructure.

Overall explanation: Hybrid cloud is the deployment model that connects on-premises infrastructure with public cloud services, allowing organizations to keep some resources local while leveraging cloud benefits for others. This approach is common when organizations face regulatory requirements, legacy application constraints, or want to gradually migrate to the cloud. The scenario describes classic hybrid cloud characteristics: distributed workloads across on-premises and Azure with integrated identity management.

Question 39

Your organization has multiple Azure subscriptions across different departments. Management requires a solution to enforce that all virtual machines must have specific tags applied (Department, CostCenter, and Owner) before they can be created. Which Azure governance feature should you implement to meet this requirement?

- **Resource locks**

- This answer is incorrect. Resource locks are used to prevent accidental deletion or modification of existing Azure resources by applying CanNotDelete or ReadOnly locks. Resource locks do not enforce tagging requirements or prevent resource creation based on compliance rules; they only protect existing resources from changes.

- **Role-based access control (RBAC)**

- This answer is incorrect. Role-based access control (RBAC) manages who has access to Azure resources, what they can do with those resources, and what areas they have access to. While RBAC controls permissions and access, it does not enforce specific resource configurations like required tags or prevent resource creation based on compliance criteria.

- **Azure Policy**

- This answer is correct. Azure Policy allows you to create, assign, and manage policies that enforce rules and effects over your resources, including requiring specific tags to be present on resources before they can be created. You can use built-in policy definitions or create custom policies to enforce tagging requirements and prevent non-compliant resources from being deployed.

- **Management groups**

- This answer is incorrect. Management groups provide a level of scope above subscriptions to efficiently manage access, policies, and compliance across multiple subscriptions. While management groups can be used to organize subscriptions and apply policies at scale, they are not the feature that directly enforces tagging requirements—Azure Policy performs that function, though it can be applied at the management group level.

Overall explanation: Azure Policy is the primary governance feature for enforcing organizational standards and assessing compliance at scale. In this scenario, Azure Policy is the correct solution because it can enforce mandatory tagging requirements and prevent the creation of non-compliant resources, ensuring that all virtual machines have the required Department, CostCenter, and Owner tags before deployment.

Question 40

A company is considering migrating their on-premises infrastructure to Azure. They currently operate 50 physical servers in their own datacenter with associated costs for hardware maintenance, electricity, cooling, and IT staff. The IT director wants to understand the potential cost savings over a 5-year period by comparing their current on-premises costs with equivalent Azure services. Which Azure tool should the IT director use to generate this comparison?

-  **Total Cost of Ownership (TCO) Calculator**
 - This answer is incorrect. The TCO Calculator was deprecated at the end of August 2025 and has since been retired.
-  **Azure Cost Management and Billing**
 - This answer is incorrect. Azure Cost Management and Billing is used to monitor, analyze, and optimize costs for resources already running in Azure. It helps track current Azure spending and usage patterns but does not perform comparisons between on-premises infrastructure costs and Azure costs for migration planning purposes.
-  **Azure Advisor**
 - This answer is incorrect. Azure Advisor provides personalized recommendations to optimize Azure deployments across cost, security, reliability, performance, and operational excellence. While it can help reduce costs for existing Azure resources, it does not compare on-premises infrastructure costs with Azure costs for migration planning.
-  **Azure Pricing Calculator**
 - This answer is correct. The Azure Pricing Calculator allows organizations to plan and forecast cloud expenses, evaluate different configurations and pricing models, and make informed decisions about service selection and deployment options. Whether in a discovery phase trying to figure out what to use and what offers to apply, or in a post-purchase phase trying to optimize the environment and see negotiated prices, the Azure Pricing Calculator fulfills both new and existing customers' needs.

Overall explanation: With the retirement of the TCO Calculator, the Azure Pricing Calculator is now the primary tool for estimating and comparing costs for planned Azure deployments, including multi-year cost projections for migration planning.

Question 41

Your organization runs multiple web applications across several Azure regions. The IT manager wants to be notified when Azure plans maintenance that might affect your company's virtual machines, and also wants to review the root cause analysis when service interruptions occur. Which Azure service should you recommend?

-  **Microsoft Defender for Cloud**
 - This answer is incorrect. Microsoft Defender for Cloud is a cloud security posture management (CSPM) and cloud workload protection platform (CWPP) that helps secure your Azure resources. While it provides security recommendations and threat protection, it does not notify you about planned Azure maintenance or provide information about Azure service health incidents.
-  **Azure Monitor**
 - This answer is incorrect. Azure Monitor is designed for collecting, analyzing, and acting on telemetry data from your Azure and on-premises resources. While it helps you track the performance and availability of your own applications and resources, it does not provide information about planned Azure platform maintenance or Azure service incidents with root cause analysis.
-  **Azure Service Health**
 - This answer is correct. Azure Service Health provides personalized alerts and guidance when Azure service issues, planned maintenance, or other changes may affect your resources. It includes Service Issues (current problems), Planned Maintenance (upcoming maintenance events), and Health Advisories, along with detailed root cause analysis (RCA) reports after incidents are resolved.
-  **Azure Advisor**
 - This answer is incorrect. Azure Advisor is a personalized cloud consultant that provides best practice recommendations to optimize your Azure deployments for cost, security, reliability, operational excellence, and performance. It does not provide notifications about planned Azure maintenance or information about Azure service disruptions and their root cause analyses.

Overall explanation: Azure Service Health is specifically designed to keep you informed about the health of Azure services and regions that matter to your organization. It provides three distinct experiences: Service Issues for active problems, Planned Maintenance for scheduled events, and Health Advisories for changes requiring action, making it the appropriate tool for tracking both planned maintenance and service incident root cause analyses.

Question 42

Your company has a production resource group named RG-Prod that contains critical virtual machines and storage accounts. The IT manager is concerned that resources might be accidentally deleted by administrators who have Contributor access to the subscription. You need to prevent the accidental deletion of resources in RG-Prod while still allowing administrators to modify resource configurations. What should you implement?

-  **Enable Microsoft Defender for Cloud on the RG-Prod resource group**
 - This answer is incorrect. Microsoft Defender for Cloud is a security posture management and threat protection service that provides security recommendations and alerts. It does not prevent users from deleting resources; instead, it focuses on identifying security vulnerabilities and threats. Resource locks, not Defender for Cloud, are the appropriate mechanism for preventing resource deletion.
-  **Apply a Read-only lock on the RG-Prod resource group**
 - This answer is incorrect. A Read-only lock would prevent both deletion and modification of resources within the resource group. Since the requirement specifies that administrators need to continue modifying resource configurations (such as changing VM sizes or updating storage account settings), a Read-only lock would be too restrictive and prevent necessary operational changes.
-  **Remove Contributor permissions and assign Reader permissions to all administrators**
 - This answer is incorrect. Removing Contributor permissions and assigning only Reader permissions would prevent accidental deletions, but it would also prevent administrators from making any modifications to resource configurations. This solution does not meet the requirement to allow administrators to continue modifying resources, making it too restrictive for operational needs.
-  **Apply a Delete lock on the RG-Prod resource group**

- This answer is correct. A Delete lock prevents resources from being deleted while still allowing administrators to read and modify existing resources. This lock type meets both requirements: it protects against accidental deletion of critical production resources while allowing Contributor-level administrators to continue making necessary configuration changes to the resources within RG-Prod.

Overall explanation: Resource locks in Azure provide a way to prevent accidental modification or deletion of critical resources. A Delete lock specifically prevents deletion operations while allowing all other operations, making it the ideal solution when you need to protect resources from accidental removal but still allow configuration changes. This is a common governance requirement in production environments where operational flexibility must be balanced with protection against accidental data loss.

Question 43

Your company is developing a new web application and wants to minimize infrastructure management while maintaining control over the application code and configuration. The development team needs a solution where they can deploy their code directly without managing virtual machines, operating systems, or underlying hardware. Which Azure service model best meets these requirements?

-  **Platform as a Service (PaaS) using Azure App Service**
 - This answer is correct. Azure App Service is a PaaS offering that allows developers to deploy application code directly without managing the underlying infrastructure, operating systems, or virtual machines. The platform handles infrastructure management, scaling, patching, and maintenance while developers retain full control over their application code and configuration settings.
-  **Infrastructure as a Service (IaaS) using Azure Virtual Machine Scale Sets**
 - This answer is incorrect. Azure Virtual Machine Scale Sets is an IaaS offering that provides automated scaling for virtual machines but still requires management of the operating systems, patches, and infrastructure. Like standard virtual machines, this solution requires significant infrastructure management effort from the development team.
-  **Infrastructure as a Service (IaaS) using Azure Virtual Machines**

- This answer is incorrect. Azure Virtual Machines is an IaaS offering that requires the development team to manage the virtual machines, operating systems, patches, and underlying infrastructure. This does not meet the requirement of minimizing infrastructure management, as the team would be responsible for maintaining the OS and runtime environment.
-  **Software as a Service (SaaS) using Microsoft 365**
 - This answer is incorrect. Microsoft 365 is a SaaS offering that provides ready-to-use applications like Outlook, Teams, and Word. With SaaS, users consume pre-built software applications and have no control over the underlying code or ability to deploy custom application code, which does not meet the scenario requirements.

Overall explanation: Platform as a Service (PaaS) is designed specifically for scenarios where developers want to focus on application development while the cloud provider manages the underlying infrastructure, operating systems, and runtime environments. Azure App Service exemplifies PaaS by providing a managed platform for web applications that eliminates infrastructure management tasks while preserving developer control over code and application settings.

Question 44

Your company is deploying a business-critical application to Azure that must remain available even if an entire Azure datacenter fails, but the application does not need to survive a regional disaster. The solution must minimize costs while meeting the availability requirement. Which redundancy option should you recommend for the Azure Storage account that will host the application data?

-  **Zone-redundant storage (ZRS)**
 - This answer is correct. Zone-redundant storage (ZRS) replicates data synchronously across three availability zones within the same region, ensuring data remains available even if an entire datacenter (availability zone) fails. This meets the requirement to survive a datacenter failure without the additional cost of geo-replication, since regional disaster recovery is not required.
-  **Locally redundant storage (LRS)**
 - This answer is incorrect. Locally redundant storage (LRS) replicates data three times within a single datacenter in one availability zone. If an entire datacenter

fails, data stored with LRS would become unavailable, which does not meet the requirement to survive a datacenter failure.

- **✗ Geo-redundant storage (GRS)**

- This answer is incorrect. Geo-redundant storage (GRS) replicates data to a secondary region hundreds of miles away, providing protection against regional disasters. While this would meet the datacenter failure requirement, it exceeds what is needed and incurs unnecessary costs since the scenario explicitly states that regional disaster recovery is not required.

- **✗ Geo-zone-redundant storage (GZRS)**

- This answer is incorrect. Geo-zone-redundant storage (GZRS) combines zone redundancy in the primary region with geo-replication to a secondary region. This provides the highest level of durability and availability but is more expensive than needed, since the scenario does not require protection against regional disasters.

Overall explanation: Azure Storage redundancy options provide different levels of protection at varying costs. Zone-redundant storage (ZRS) is the most cost-effective solution that protects against datacenter-level failures by replicating across availability zones within a region, while LRS only protects against drive/rack failures, and GRS/GZRS add unnecessary regional protection for this scenario.

Question 45

A retail company wants to build a mobile application that allows customers to create accounts, sign in using their social media credentials (Facebook, Google, or email), and manage their shopping profiles. The company needs an Azure identity service that can handle customer authentication without requiring customers to have corporate credentials. Which Azure service should the company implement?

- **✓ Microsoft Entra ID B2C (Business-to-Consumer)**

- This answer is correct. Microsoft Entra ID B2C is specifically designed for customer-facing applications where consumers can sign up and sign in using their preferred social, enterprise, or local account identities. It supports integration with social identity providers like Facebook and Google, and allows customers to create and manage their own accounts without needing corporate or organizational credentials.

- **✗ Microsoft Entra Domain Services**
 - This answer is incorrect. Microsoft Entra Domain Services provides managed domain services such as domain join, group policy, and LDAP for Azure virtual machines, primarily for lift-and-shift scenarios of legacy applications. It is not designed for customer identity and access management in consumer-facing applications.
- **✗ Microsoft Entra ID (standard tenant)**
 - This answer is incorrect. While Microsoft Entra ID (standard tenant) provides identity and access management, it is designed for employees and organizational users within an enterprise. It does not provide the consumer-focused features like social identity provider integration and self-service customer account management that are essential for customer-facing applications.
- **✗ Microsoft Entra ID B2B (Business-to-Business)**
 - This answer is incorrect. Microsoft Entra ID B2B is designed for collaboration between business partners and external organizations, allowing guest users from other companies to access your resources. It is not intended for consumer-facing applications where individual customers need to create public accounts with social identity providers.

Overall explanation: Microsoft Entra ID B2C is purpose-built for business-to-consumer scenarios where companies need to provide identity management for their customers. It enables seamless customer sign-up and sign-in experiences with support for social identity providers, customizable user interfaces, and scalability to handle millions of consumer identities, making it the ideal choice for customer-facing applications.

Question 46

Your company has deployed a web application to Azure App Service. The development team reports that users are experiencing slow page load times, but they cannot determine which specific components are causing the performance issues. You need to recommend an Azure monitoring solution that can automatically detect performance anomalies, track user behavior, and identify which dependencies are causing slowdowns. Which Azure service should you recommend?

- **✗ Azure Service Health**

- This answer is incorrect. Azure Service Health provides information about the status of Azure services and planned maintenance that might affect your resources. It focuses on Azure platform health and service incidents rather than monitoring the performance and behavior of your specific applications, so it would not help identify performance issues within the web application itself.
-  **Azure Monitor Logs**
 - This answer is incorrect. While Azure Monitor Logs can collect and analyze log data from various sources, it does not provide the specialized application performance monitoring (APM) capabilities needed for this scenario. Azure Monitor Logs focuses on log queries and analysis rather than automatic performance anomaly detection, dependency tracking, and user behavior analysis specific to applications.
-  **Microsoft Defender for Cloud**
 - This answer is incorrect. Microsoft Defender for Cloud is a security management tool that provides threat protection and security posture management for Azure resources. While it monitors security-related issues and vulnerabilities, it does not provide application performance monitoring, dependency tracking, or user behavior analysis needed to diagnose slow page load times.
-  **Application Insights**
 - This answer is correct. Application Insights is a feature of Azure Monitor specifically designed for application performance management (APM) that automatically detects performance anomalies, tracks dependencies, monitors user behavior, and provides detailed telemetry about application performance. It can identify slow dependencies, failed requests, and performance bottlenecks in web applications, making it the ideal solution for this scenario.

Overall explanation: Application Insights is Azure's application performance management (APM) service that provides comprehensive monitoring for web applications including automatic anomaly detection, dependency tracking, performance metrics, and user analytics. For this scenario requiring identification of performance bottlenecks and slow dependencies in a web application, Application Insights is the appropriate Azure service that provides all the necessary capabilities out of the box.

Question 47

Your organization needs to ensure that all Azure virtual machines deployed across multiple subscriptions are created only in specific Azure regions (East US and West Europe) to comply with data residency requirements. What should you use to enforce this requirement?

-  **Azure Policy**
 - This answer is correct. Azure Policy is designed to enforce organizational standards and assess compliance at scale. You can create a policy definition that restricts which Azure regions resources can be deployed to, and assign it across multiple subscriptions to ensure all virtual machines comply with the data residency requirements.
-  **Role-based access control (RBAC)**
 - This answer is incorrect. Role-based access control (RBAC) manages who can access Azure resources and what actions they can perform (read, write, delete, etc.). While RBAC controls permissions, it does not enforce rules about resource configurations or deployment locations.
-  **Resource locks**
 - This answer is incorrect. Resource locks prevent resources from being accidentally deleted or modified, but they do not enforce rules about where resources can be created or what configurations they must have. Resource locks are applied to existing resources to protect them, not to enforce deployment standards.
-  **Microsoft Defender for Cloud**
 - This answer is incorrect. Microsoft Defender for Cloud is a security posture management and threat protection service that helps identify security vulnerabilities and protect against threats. While it can assess compliance with security standards, it does not enforce deployment restrictions such as allowed regions for resource creation.

Overall explanation: Azure Policy is the primary tool for enforcing organizational standards and governance requirements across Azure resources. It allows you to create rules that evaluate resource properties and can deny deployments that don't meet specified criteria, such as restricting resources to specific regions. This makes Azure Policy the appropriate solution for ensuring compliance with data residency requirements.

Question 48

A manufacturing company has servers running in their on-premises datacenter, virtual machines running in AWS, and resources deployed in Azure. The IT director wants to use Azure management tools, apply Azure policies, and view all resources through a single control plane regardless of where they are physically located. Which Azure service should the company implement?

-  **Azure Arc**
 - This answer is correct. Azure Arc extends Azure management capabilities to resources outside of Azure, including on-premises servers and resources in other cloud platforms like AWS and Google Cloud. It allows organizations to manage hybrid and multi-cloud environments through a single control plane, apply Azure policies, and use Azure management tools across all resources regardless of their location.
-  **Azure Resource Manager**
 - This answer is incorrect. While Azure Resource Manager (ARM) is the deployment and management service for Azure, it natively manages only resources that are deployed within Azure itself. ARM does not extend Azure management capabilities to resources running on-premises or in other cloud platforms like AWS without additional services.
-  **Microsoft Defender for Cloud**
 - This answer is incorrect. Microsoft Defender for Cloud is a security posture management and threat protection service that can work across multi-cloud environments. However, it focuses specifically on security rather than providing the comprehensive management control plane, policy application, and unified resource management capabilities described in the scenario.
-  **Azure Monitor**
 - This answer is incorrect. Azure Monitor is a comprehensive monitoring solution for collecting, analyzing, and acting on telemetry from cloud and on-premises environments. While Azure Monitor can collect data from various sources, it does not provide the unified management control plane or the ability to apply Azure governance and policies to non-Azure resources that the scenario requires.

Overall explanation: Azure Arc is specifically designed to extend Azure's management and governance capabilities to resources outside of Azure, enabling hybrid and multi-cloud management. It projects non-Azure resources into Azure Resource Manager, allowing organizations to manage servers, Kubernetes clusters, and data services across on-premises, multi-cloud, and edge environments through a single pane of glass. This makes Azure Arc the appropriate solution for managing the company's resources distributed across on-premises datacenters, AWS, and Azure.

Question 49

A retail company runs a critical e-commerce application that must remain available to customers even if an Azure datacenter experiences an outage. The company wants to leverage Azure's infrastructure to ensure the application continues operating during such failures. Which Azure reliability benefit directly addresses this requirement?

-  **Azure's pay-as-you-go pricing model that reduces upfront infrastructure costs**
 - This answer is incorrect. The pay-as-you-go pricing model is a cost management benefit that provides financial flexibility. It does not contribute to application reliability or protect against datacenter failures, as it relates to billing rather than infrastructure resilience.
-  **Azure's support for automatic horizontal scaling to handle increased traffic loads**
 - This answer is incorrect. While automatic horizontal scaling improves performance and handles varying workloads, it relates to scalability rather than reliability. Scaling does not specifically protect against datacenter failures or ensure continuous operation during infrastructure outages.
-  **Azure's ability to deploy resources across multiple availability zones within a region**
 - This answer is correct. Availability zones are physically separate datacenters within an Azure region, each with independent power, cooling, and networking. Deploying resources across multiple availability zones ensures that if one datacenter fails, the application continues running in the other zones, directly providing the reliability and resilience needed to maintain availability during datacenter outages.

-  **Azure's compliance certifications that meet industry regulatory standards**
 - This answer is incorrect. Compliance certifications demonstrate that Azure meets various regulatory and industry standards, which relates to governance and compliance benefits. These certifications do not provide protection against datacenter outages or ensure application availability during failures.

Overall explanation: Reliability in Azure focuses on the ability of applications to remain available and resilient in the face of failures. Availability zones are a key reliability feature that provides datacenter-level fault tolerance by distributing resources across physically isolated locations within a region, ensuring business continuity even during infrastructure failures.

Question 50

Your organization stores customer data across multiple Azure services including Azure SQL Database, Azure Storage accounts, and Azure Data Lake Storage. The compliance team needs to discover where sensitive personal information resides, classify data according to regulatory requirements, and track how data is being accessed across all these data sources from a central location. Which Azure service should you recommend?

-  **Microsoft Purview**
 - This answer is correct. Microsoft Purview is a unified data governance service that provides data discovery, classification, and lineage capabilities across an organization's data estate. It can automatically scan and classify sensitive data across Azure services, apply data sensitivity labels, and provide insights into how data is accessed and used, which directly addresses the compliance team's requirements.
-  **Azure Policy**
 - This answer is incorrect. Azure Policy is used to enforce organizational standards and assess compliance at scale by creating, assigning, and managing policies for Azure resources. While it helps ensure resources are configured correctly, it does not provide data discovery, classification, or tracking of sensitive information within data stores.
-  **Microsoft Defender for Cloud**
 - This answer is incorrect. Microsoft Defender for Cloud is primarily focused on security posture management and threat protection for cloud workloads. While it

helps secure resources and identify security vulnerabilities, it does not provide unified data governance, data discovery, classification, or tracking of sensitive data across data estates.

- **✗ Microsoft Entra ID**

- This answer is incorrect. Microsoft Entra ID is an identity and access management service that handles authentication and authorization for users and applications. While it controls who can access resources, it does not discover, classify, or track sensitive data within those resources or provide data governance capabilities.

Overall explanation: Microsoft Purview is designed specifically for unified data governance across an organization's data estate. It provides automated data discovery, sensitive data classification, data lineage tracking, and compliance reporting capabilities that enable organizations to understand where their sensitive data resides and how it's being used across multiple data sources.
